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EACL 2026 RABAT · MOROCCO March · Mars · 2026 · مارس

SCAN: Semantic Document Layout Analysis for Textual and Visual Retrieval-Augmented Generation

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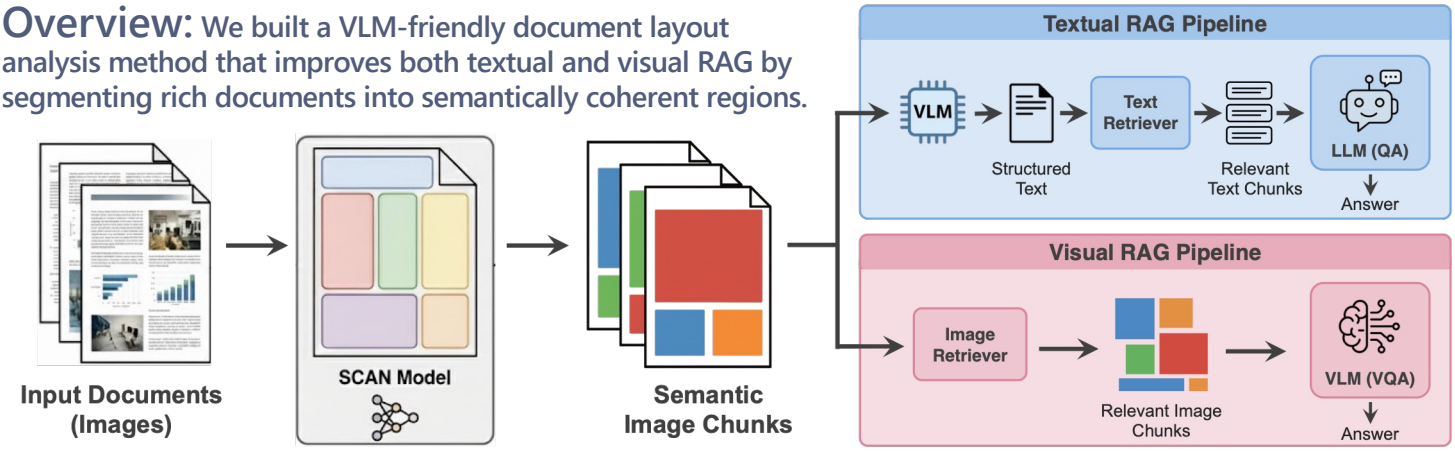
Overview: We built a VLM-friendly document layout analysis method that improves both textual and visual RAG by segmenting rich documents into semantically coherent regions.



01 Background: Textual/Visual RAG

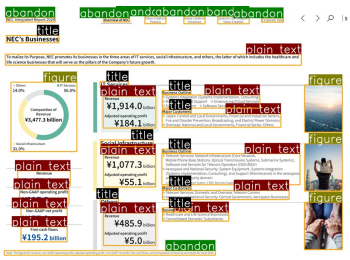
- **Textual RAG** first converts visual documents into texts using a VLM, followed by a usual RAG pipeline.
- **Visual RAG** retrieves images of pages and uses a VLM to answer the query.

Challenge: VLMs cannot accurately process an entire rich document page.

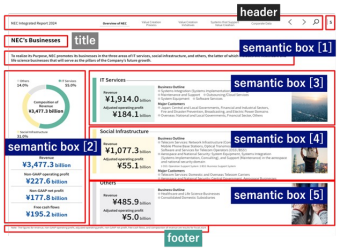
▶ We need to split pages to reduce the VLMs' burden.

02 Method: Semantic Layout Analysis

- Existing document layout analysis splits pages into too fine-grained chunks, which is not optimal for VLMs.
- We propose **semantic document layout analysis** and annotate 24k+ pages with semantic layout labels.
- We fine-tune an object detection model with the labels to build the SCAN model.

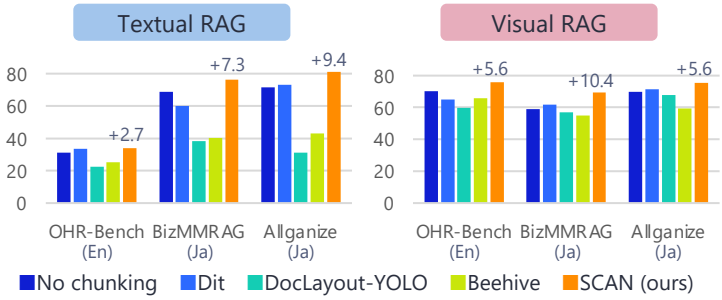


Conventional layout analysis result

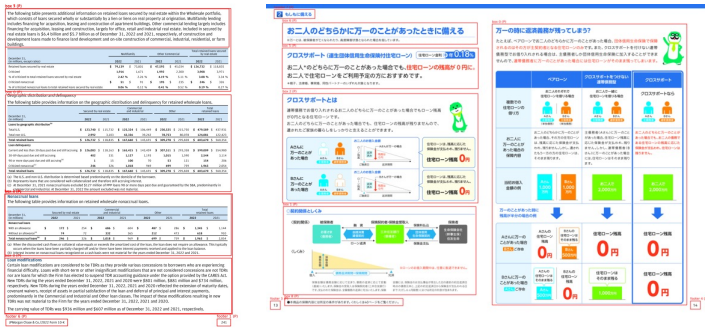


Our semantic layout analysis result

03 Experiment on RAG Benchmarks



04 Case Analysis



05 Cost Comparison of VLM Text Conv.

- We sampled 10 images randomly and used Qwen2.5-VL 72B for text conversion.
- Despite the increase in token counts, SCAN takes less time due to reduced attention computation.

Setting	# Input Tokens	# Output Tokens	# Chunks	# Input Tokens per Chunk	Time (s)
No chunking	1320.4	991.9	1.0	1320.4	68.0
SCAN (ours)	9683.1	2515.0	12.4	**780.9**	**56.3**